



MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 3 UNIT 1
Lesson Video 1 Transcript

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PAUL MCDONAGH-SMITH: Welcome to Module 3. Over the first two modules, you've built a substantial foundation. You can define agentic AI, you understand how agents work, you've explored real-world workflows. Now comes the question that separates experimentation from enterprise impact: How do you scale?

One of the most persistent risks in enterprise AI right now is what I'd call the seduction of the demo. These models make it extraordinarily easy to build an impressive demo in a short period of time. You can prototype a workflow in a day or two that looks like it solves a real business problem. The demo works, stakeholders are excited, resources are allocated, and then deployment begins. And then is where you discover the corner cases. The edge conditions where the model hallucinates, selects the wrong tool, misinterprets the intent, or propagates an early error through every subsequent step.

The distance between "this demo is amazing" and "this is ready for production at scale" is often greater than organizations anticipate. And closing, narrowing that distance isn't a technical debugging exercise. It's a design challenge that involves governance, measurement, human oversight, and organizational readiness.

Reframing what "scaling" means

MCDONAGH-SMITH: When most leaders hear "scale," they think take what works and make it bigger. More users, more transactions, more scope. With agentic AI, that model isn't completely adequate. Scaling is not enlarging a pilot. It's transitioning from a controlled environment where a small team understands every decision the agent makes to an operational context where the agent acts across functions, across data sources, and across situations that no one has fully explicitly tested. That transition introduces qualitatively new challenges. The failure modes change, the governance requirements intensify, the measurement problem deepens, and the organizational dynamics shift, because scaling means the technology now touches people who may not have volunteered for the pilot.

The research is consistent on one point: ambitious monolithic initiatives, let's build one agent that does everything, tend to underperform relative to approaches that start small, iterate rapidly, and scale through a portfolio of focused experiments. Not because ambition is wrong, but because the technology rewards a particular kind of strategic discipline. Fast learning loops, bounded scope, continuous measurements, and the willingness to stop scaling what isn't working.

Conclusion

MCDONAGH-SMITH: So, as we enter this module, I'd like you to calibrate, recalibrate your expectations accordingly. We're not going to tell you that scaling is easy, because frankly, often it isn't. We're going to try to equip you with the frameworks for doing it honestly, for assessing risk, designing governance, sequencing deployment, and building the organizational muscle to sustain it. In the next video, we'll look at how to design for uncertainty. The practical architectures of trust, safeguards, and autonomy calibration that separates a resilient scaling strategy from an interconnected series of pilots.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.